

PGR207

Deep Learning

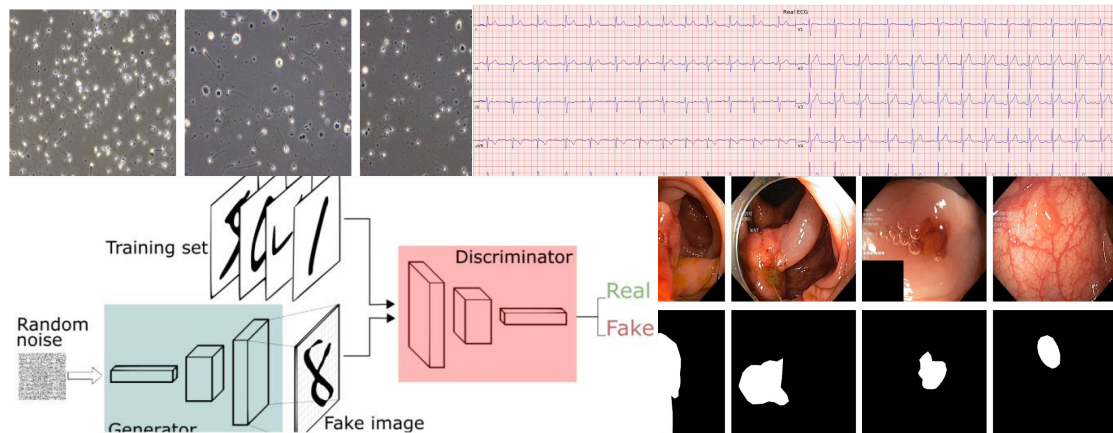
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Introduction to Deep Learning



Who am I?

simulamnet



Level?

The best
communication
method?

Your knowledge?

Who are you?

Resources?

Your capacity?

Batch
representative?

class representatives?

1 or 2? One Female and One Male?

Name?

Email?

Outline

- Introduction to course content
- Course evaluation method, and exams
- What is deep learning?
- Application of Deep Learning
- Deep Learning Frameworks
- Installing a Deep Learning framework
- Using Google collab (Kaggle platform)
- Jupyter notebooks

Introduction to course content

Workload 🐒

Participation in lectures and exercises - **48 hours (12 weeks x 4 hrs)**

Self-study - **72 hours**

Execution of and preparation for the exam - **80 hours**

Evaluation -

- One individual project - Mandatory

- Paper + Code (Notebooks)

- Final group project (groups -2-3) - Mandatory

- Draft paper (report) + Code (Notebooks)
 - Grades - A-F

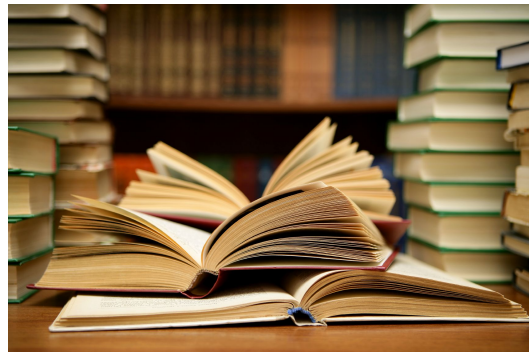


References

- **Machine Learning with Pytorch and Scikit-Learn: Develop machine learning and deep learning models with Python. Sebastian Raschka (Author), Yuxi (Hayden) Liu (Author), Vahid Mirjalili (Author)**

[From Chapter 11]

- Mastering PyTorch, Ashish Ranjan Jha (Author)
- <https://pytorch.org/> (Deep learning framework) - Online



To learn

Section	Description	Week number (total number of weeks = 12)
Introduction to DL	Introduction to course content, course evaluation method, and exams, What is deep learning? Application of Deep Learning?, Deep Learning Frameworks, Installing a Deep learning framework, Using Google collab, Jupyter notebooks	Week 1
Basics of Neural networks	Artificial neurons and neural networks, implementing and training multilayer neural network from scratch.	Week 2
Pytorch for DL	What is Pytorch and Why Pytorch? Tensors for data representation, Pytorch DL pipeline, ANN using Pytorch,	Week 3
Deep Convolutional Neural Networks	What is a CNN? Where can we use CNNs? Identifying differences in Classification, Detection, and Segmentation. Implementing CNN models using Pytorch.	Weeks 4, 5

To learn

Section	Description	Week number (total number of weeks = 12)
Recurrent Neural Networks and Transformers	What is an RNN and where can we use RNNs? Understanding RNN and the mechanism. Implementing an RNN model using Pytorch. What is a transformer? Understanding the original transformer network.	Weeks 6, 7
Generative Adversarial Networks	Introduction to generative models, Autoencoders. Implementing a GAN model using Pytorch in Google collab. Improving the quality of synthetic data generated using GANs. Different types of GANs and applications.	Weeks 8,9
Graph Neural Networks / LLMs	Introduction to graph data and GNN and LLMs	Week 10
Deep Reinforcement learning	What is reinforcement learning and why is it? Basics of RL algorithms, implementing RL algorithms. Implementing deep Q-learning using Pytorch.	Week 11, 12

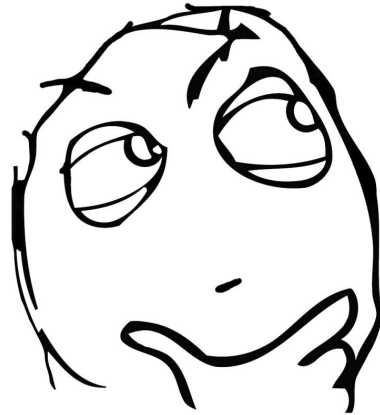
What is deep learning?

What do you think about these photos?



A brain riding a rocket
ship heading towards the
moon.

A dragon fruit wearing karate
belt in the snow.



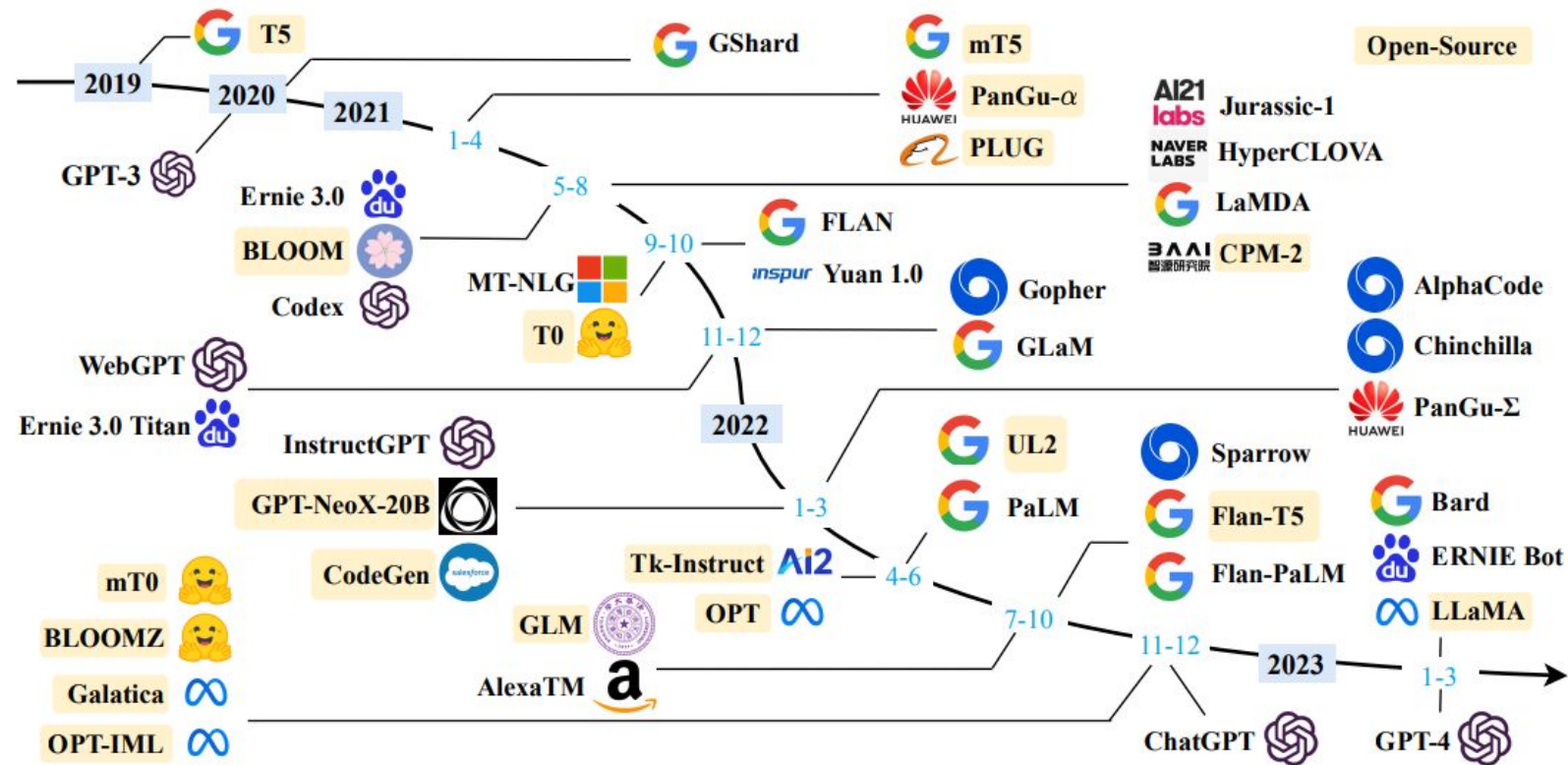
A brain riding a rocket ship heading towards the moon.



A dragon fruit wearing karate belt in the snow.



LLMs



<https://openai.com/index/dall-e-3/>

Can we use DL knowledge to earn money?
(in addition to doing a job in this path)

<https://www.kaggle.com/>

<https://grand-challenge.org/challenges/>

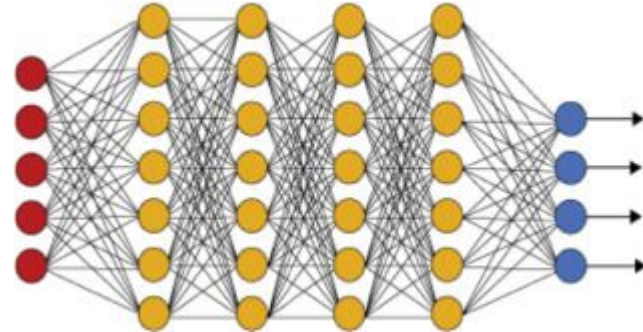
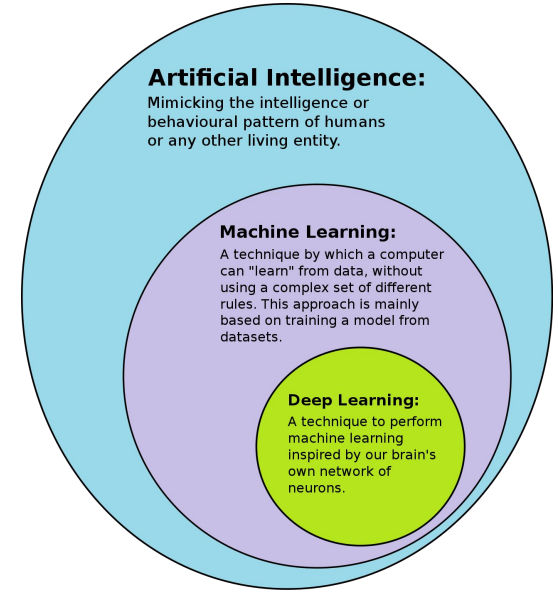
<https://www.aicrowd.com/>

<https://www.topcoder.com/>

Definition of DL

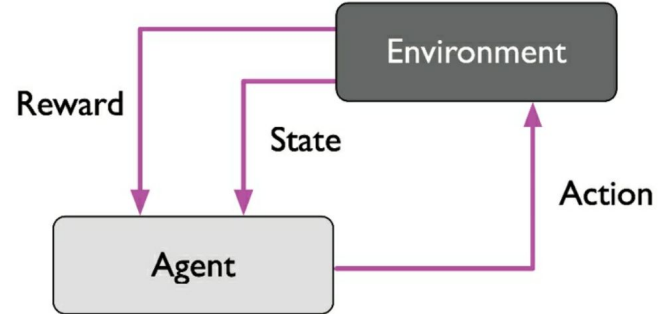
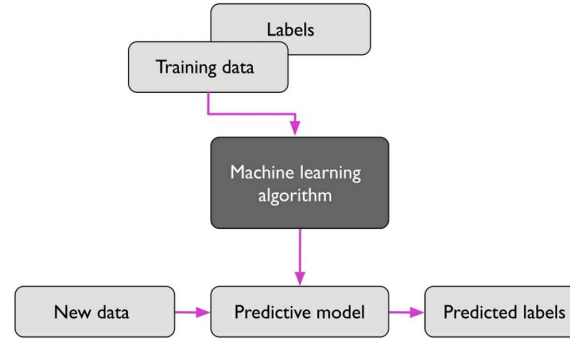
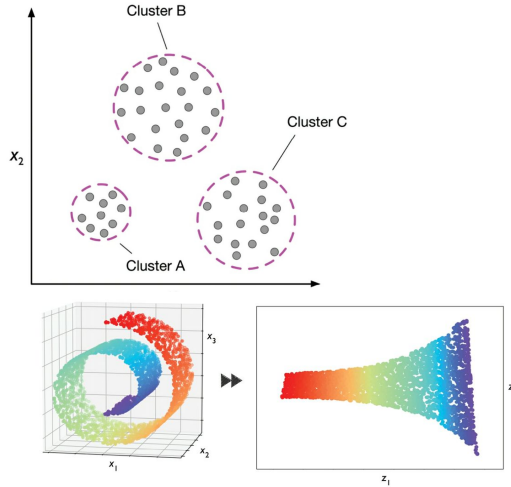
Deep learning is a subset of machine learning, which is essentially a neural network with three or more layers. (DNN)

These neural networks attempt to simulate the behavior of the human brain - allowing it to “learn” from large amounts of data.



Three different types of machine learning

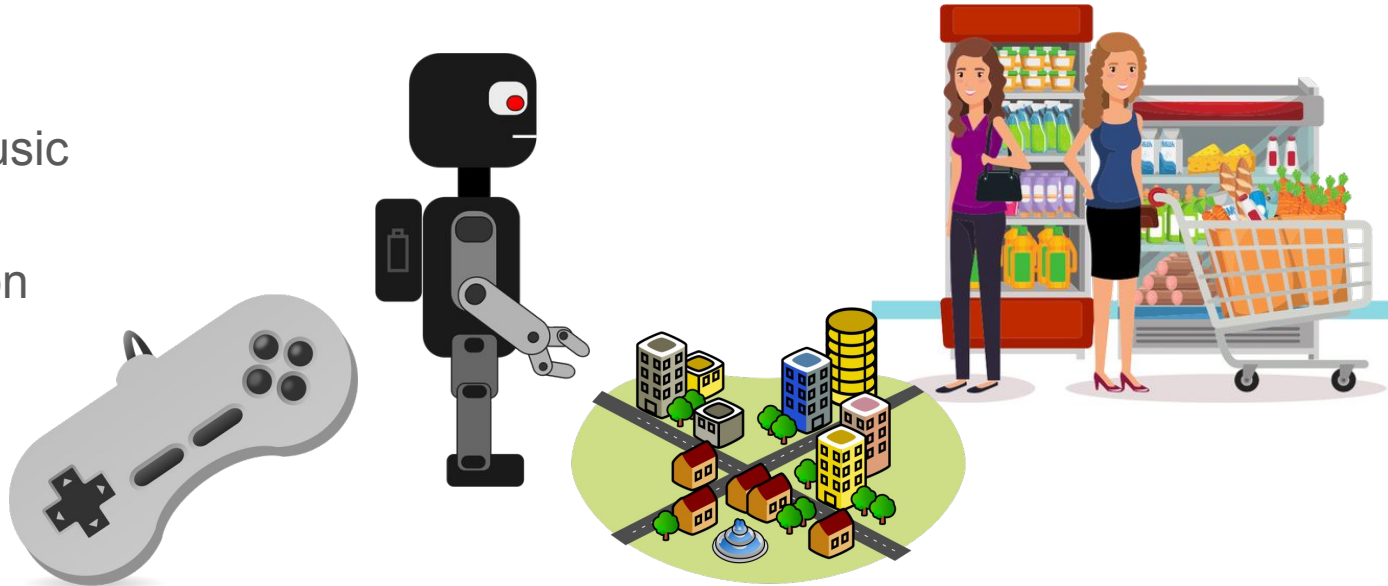
- Supervised learning
- Unsupervised learning
- Reinforcement learning



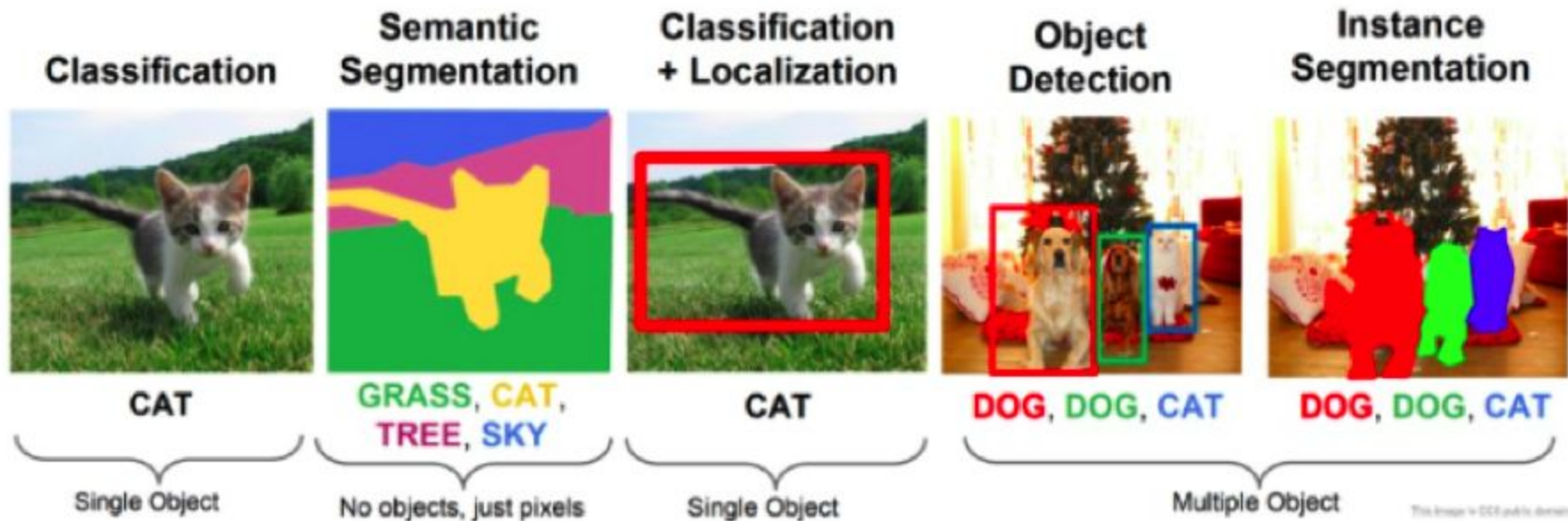
Application of Deep Learning

Some applications of DL

- Computer vision
- Natural language processing
- Self Driving Cars or Autonomous Vehicles (more than computer vision)
- Healthcare
- Entertainment
- Composing Music
- Robotics
- Fraud Detection
- E-commerce



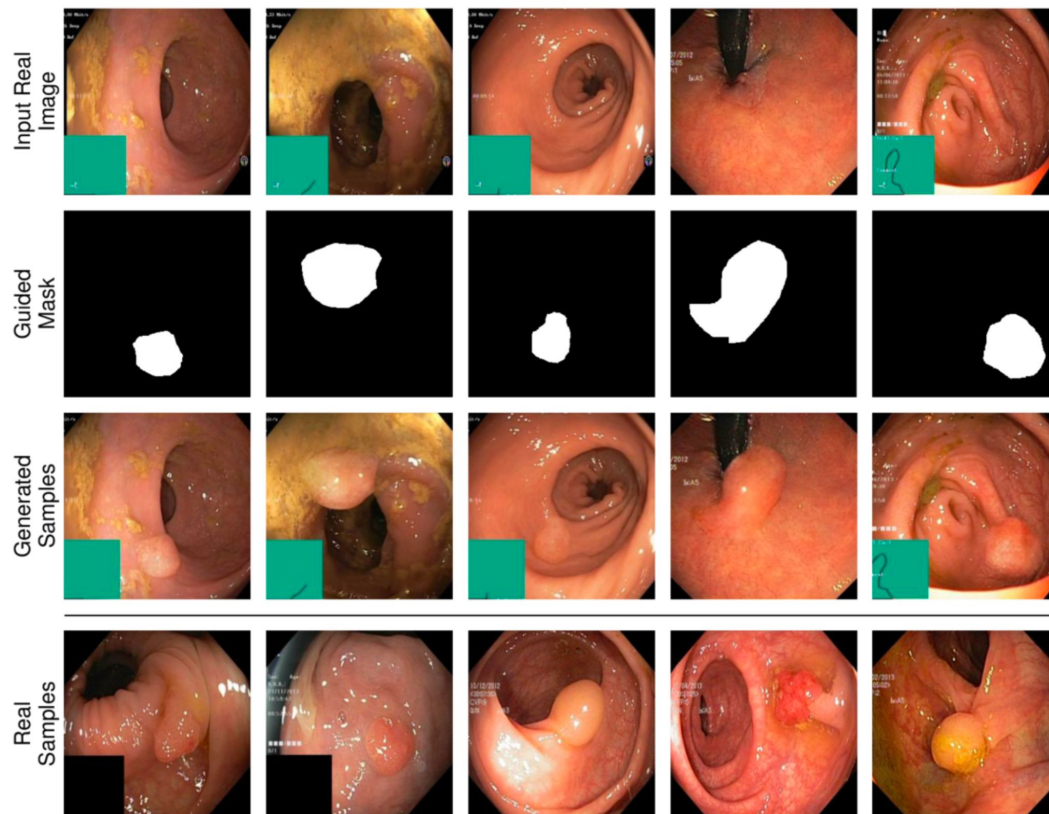
Computer vision categories



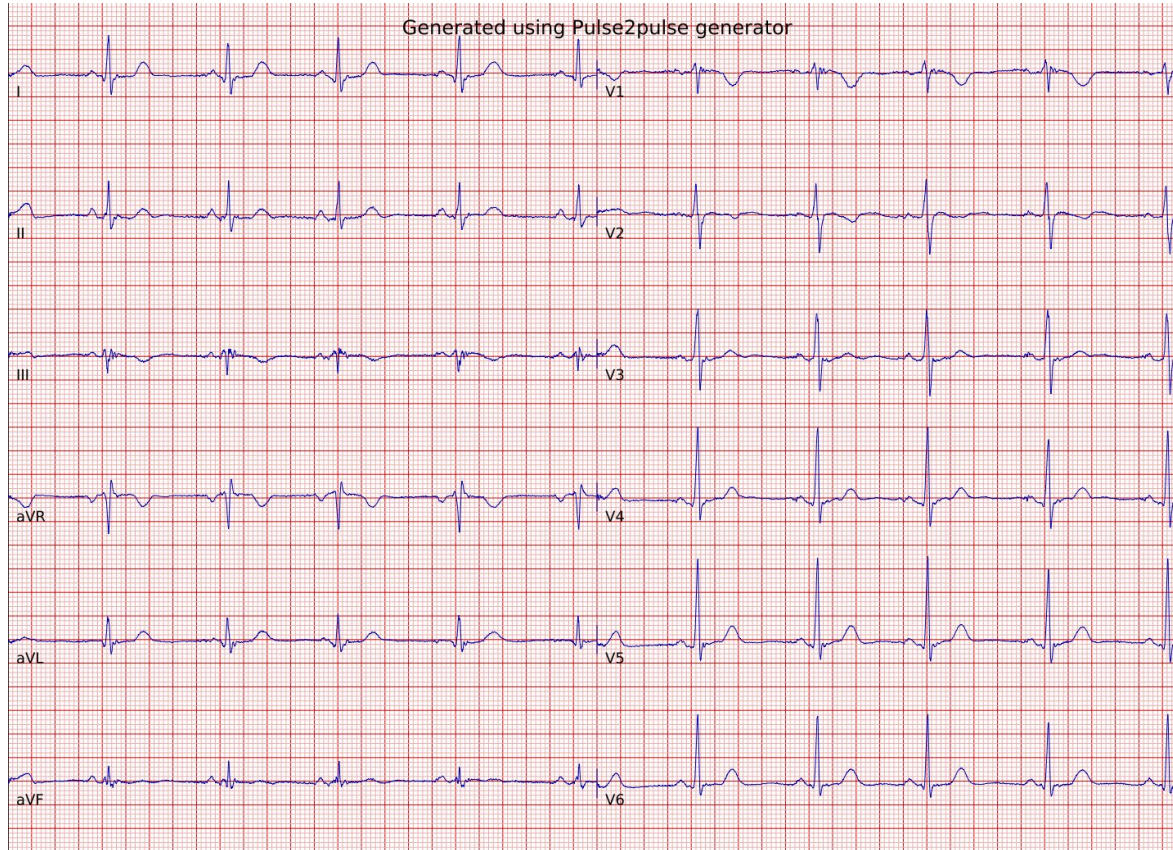
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Our projects?

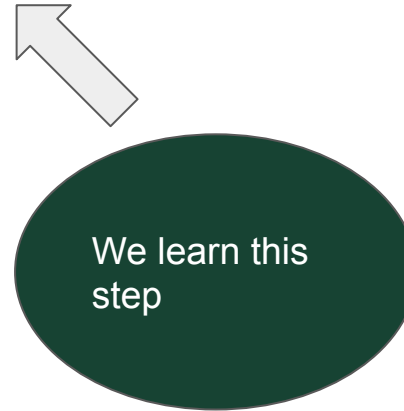
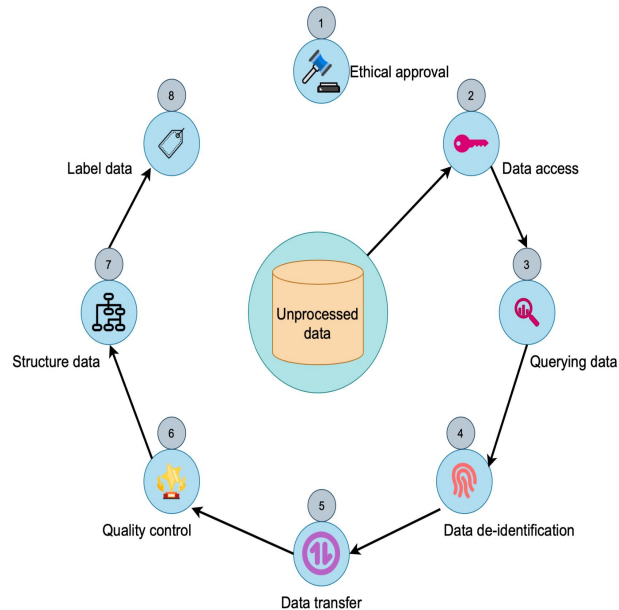
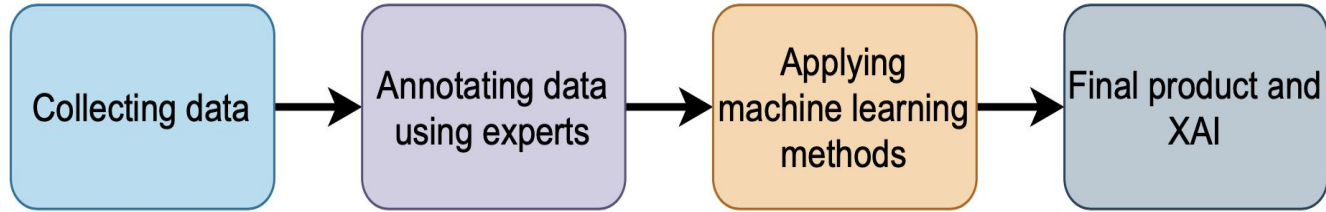
Synthetic polyp generation



Synthetic ECG



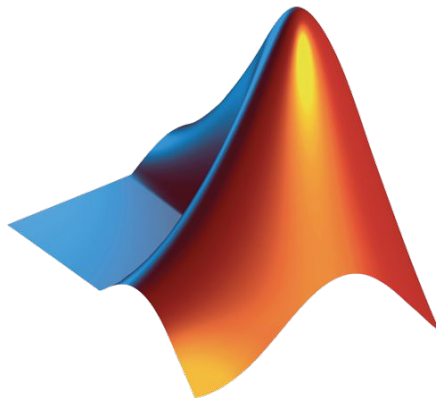
The main pipeline



Deep Learning Frameworks



TensorFlow



<https://pytorch.org/>

Hardware



CPU



GPU



TPU



Practical Session (Week 1)